$$\dot{\underline{X}} = F(\underline{X}, t)$$
$$\underline{X} = \left[\underline{X}_X, \underline{X}_Y, \underline{X}_Z, \dot{\underline{X}}_X, \dot{\underline{X}}_Y, \dot{\underline{X}}_Z, \beta\right]^\top$$
$$\dot{X} = F(X, t)$$
$$X = \left[X_1, X_2, X_3, \dot{X}_1, \dot{X}_2, \dot{X}_3, \gamma\right]^\top$$
$$x(t) = F\left[t, t_0, x_0(t)\right]$$
$$\dot{x} = f(x, t) = \left[f_1, f_2, f_3, f_4, f_5, f_6, f_7\right]^\top$$
$$\dot{X}(t) = \dot{X}^*(t) + \left[\frac{\partial \dot{X}(t)}{\partial X(t)}\right]^* \left[X(t) - X^*(t)\right] + \cdots$$